Student Performance Analysis and Prediction Using Exploratory Data Analysis and Deep Learning

U21ADP05 Exploratory Data Analysis and Visualization

REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **"Student Performance Analysis and Prediction Using Exploratory Data Analysis and Deep Learning"** is the bonafide work of **"Shreenidhi E (24CS197)"** who carried out the project work under my supervision.

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ABSTRACT

Student academic performance is influenced by various factors, including study habits, family background, social conditions and previous academic results. Analyzing these factors helps educational institutions understand student performance and identify patterns that influence academic achievement. This project, “Student Performance Analysis and Prediction Using Exploratory Data Analysis and Deep Learning,” aims to analyze student academic data and predict final examination grades using data analysis and predictive modeling techniques.

The project uses the Student Performance dataset obtained from the UCI Machine Learning Repository, consisting of 395 student records and 33 attributes. Data preprocessing techniques such as missing value analysis, duplicate checking, outlier detection, categorical encoding and numerical scaling are performed to prepare the dataset for analysis. Exploratory Data Analysis (EDA) is carried out using statistical measures and visualizations, including histograms, boxplots, bar charts, scatter plots, correlation heatmaps and pairplots. The seven stages of data visualization are also implemented to represent meaningful patterns and relationships in the dataset.

A Multi-Layer Perceptron (MLP) model is developed using TensorFlow/Keras to predict students' final mathematics grades. Hyperparameter tuning is performed to identify a suitable model configuration. The MLP model is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared (R^2) metrics and compared with Linear Regression and Random Forest models. In the latest evaluation, the MLP achieved an MAE of 1.8415 and an R^2 score of 0.7302, while Random Forest achieved the best overall performance with an R^2 score of 0.8280.

Finally, an interactive dashboard is developed using Streamlit to provide data exploration, visualization, model evaluation and student performance prediction. The project demonstrates how Exploratory Data Analysis, data visualization and Deep Learning techniques can be applied to educational data to support better understanding of student academic performance.

Keywords: Student Performance, Exploratory Data Analysis, Data Visualization, Deep Learning, Multi-Layer Perceptron, Streamlit, Student Grade Prediction.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1	EDA	Exploratory Data Analysis
2	MLP	Multi-Layer Perceptron
3	AI	Artificial Intelligence
4	ML	Machine Learning
5	DL	Deep Learning
6	UCI	University of California, Irvine
7	CSV	Comma-Separated Values
8	G1	First Period Grade
9	G2	Second Period Grade
10	G3	Final Period Grade
11	IQR	Interquartile Range
12	MAE	Mean Absolute Error

CHAPTER 1

INTRODUCTION

1.1 Introduction

Education plays an important role in shaping students' academic and professional development. Student performance is influenced by several factors, including study habits, family background, social conditions, previous academic results and personal lifestyle.

Analyzing student academic data helps educational institutions understand the factors that influence learning outcomes. Exploratory Data Analysis (EDA) is used to examine datasets, identify patterns, discover relationships and understand the distribution of data. Deep Learning techniques can also be applied to develop predictive models that estimate students' academic performance.

This project focuses on analyzing and predicting student performance using the Student Performance dataset obtained from the UCI Machine Learning Repository. The project uses Python, data visualization techniques and a Multi-Layer Perceptron (MLP) model to study academic performance and generate predictions.

1.2 Problem Statement

Traditional methods of evaluating student performance mainly depend on examination marks and manual analysis. These methods may not identify important relationships between academic performance and other factors.

Therefore, an automated system is required to analyze student-related information, visualize important patterns and predict academic performance using machine learning and deep learning techniques.

1.3 Objectives

The main objectives of this project are:

1. To acquire and understand a real-world student performance dataset.
2. To perform data cleaning and preprocessing.
3. To analyze student performance using statistical techniques.
4. To create meaningful visualizations to identify patterns and relationships.

5. To implement the seven stages of data visualization.
6. To develop a Deep Learning model using a Multi-Layer Perceptron.
7. To evaluate the model using suitable performance metrics.
8. To develop an interactive dashboard for student performance analysis and prediction.

1.4 Scope of the Project

The project focuses on analyzing students' academic performance using demographic, social, family and academic-related attributes. It includes data preprocessing, exploratory analysis, visualization, model development and performance evaluation.

The system provides an interactive dashboard through which users can explore student data and obtain predicted academic performance.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

Study 1: Cortez and Silva (2008)

Cortez and Silva studied the prediction of secondary school student performance using data mining techniques. Their research considered demographic, social and school-related factors to understand student achievement. This work provides the foundation for the dataset used in this project.

Study 2: Dua and Graff (2017)

Dua and Graff presented the UCI Machine Learning Repository as a collection of datasets for machine learning research and experimentation. The Student Performance dataset used in this project is available through this repository.

Study 3: Abadi et al. (2016)

Abadi and co-authors introduced TensorFlow, an open-source framework for numerical computation and machine learning. TensorFlow supports the development and training of neural network models.

Study 4: Pedregosa et al. (2011)

Pedregosa and co-authors introduced Scikit-learn, a Python library that provides tools for data preprocessing, machine learning algorithms and model evaluation.

Study 5: Romero and Ventura (2010)

Romero and Ventura reviewed educational data mining techniques and their applications in educational environments. Their work highlights the importance of analyzing educational data to understand student learning and performance.

2.2 Proposed System

The proposed system combines Exploratory Data Analysis, data visualization and Deep Learning to analyze student academic performance.

The system follows these major steps:

1. Dataset acquisition.
2. Data preprocessing.
3. Exploratory Data Analysis.
4. Data visualization.
5. Deep Learning model development.
6. Model evaluation.
7. Interactive dashboard development.

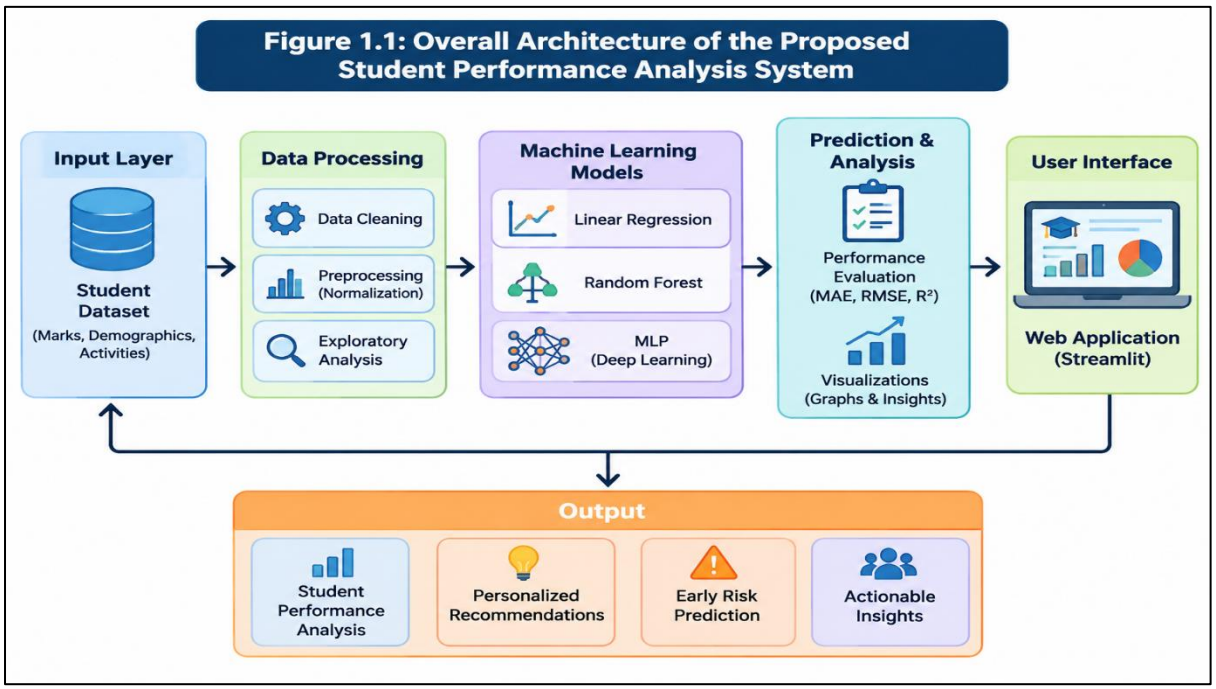


Figure 2.1 - Overall Proposed System Architecture

CHAPTER 3

DATASET DESCRIPTION AND DATA PREPROCESSING

3.1 Dataset Description

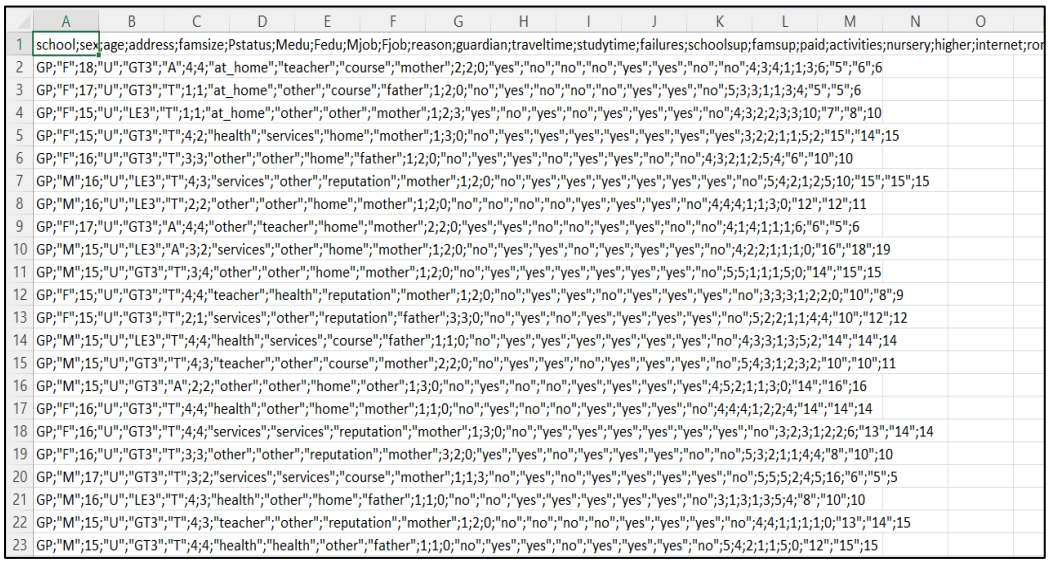
The Student Performance dataset is collected from the UCI Machine Learning Repository. The dataset was originally presented by Cortez and Silva in 2008.

This project uses the mathematics student dataset named student-mat.csv.

The dataset contains:

- Total records: 395 students.
- Total attributes: 33.
- Target variable: G3.
- Target variable range: 0 to 20.

The G3 attribute represents the final mathematics examination grade.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	school;sex;age;address;famsize;Pstatus;Medu;Fedu;Mjob;Fjob;reason;guardian;traveltime;studytime;failures;schoolsup;famsup;paid;activities;nursery;higher;internet;ror														
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20	GP;"M";17;"U";"GT3";"T";3;2;"services";"services";"course";"mother";1;1;3;"no";"yes";"no";"yes";"yes";"yes";"yes";"yes";"no";5;5;5;2;4;5;16;"6";"5";5														
21	GP;"M";16;"U";"LE3";"T";4;3;"health";"other";"home";"father";1;1;0;"no";"no";"yes";"yes";"yes";"yes";"yes";"yes";"no";3;1;3;1;3;5;4;"8";"10";10														
22	GP;"M";15;"U";"GT3";"T";4;3;"teacher";"other";"reputation";"mother";1;2;0;"no";"no";"no";"no";"no";"yes";"yes";"yes";"yes";"no";4;4;1;1;1;0;"13";"14";15														
23	GP;"M";15;"U";"GT3";"T";4;4;"health";"health";"other";"father";1;1;0;"no";"yes";"yes";"no";"yes";"yes";"yes";"yes";"no";5;4;2;1;1;5;0;"12";"15";15														

Figure 3.1: Student Performance Dataset

Attribute	Description
Dataset Name	Student Performance - Mathematics
Source	UCI Machine Learning Repository
File Name	student -mat.csv
Total Records	395
Total Attributes	33
Target Variable	G3
Target Variable Range	0-20
Data Type	Numerical and Categorical

Table 3.1: Summary of the Student Performance Dataset.

3.2 Description of Dataset Attributes

The dataset contains different categories of attributes.

1. Demographic Attributes

- School
- Sex
- Age
- Address

2. Family and Social Attributes

- Family size
- Parent's cohabitation status
- Mother's education
- Father's education
- Mother's job
- Father's job
- Guardian
- Relationship quality

3. Academic Attributes

- Study time
- Previous failures
- School support
- Family educational support
- Extra educational support
- Paid classes
- Activities
- Nursery education
- Higher education aspiration
- Internet access

4. Lifestyle and Health Attributes

- Free time
- Going out
- Workday alcohol consumption
- Weekend alcohol consumption
- Health

5. Academic Grade Attributes

- G1 – First period grade.
- G2 – Second period grade.
- G3 – Final grade.

3.3 Data Preprocessing

Data preprocessing is an important step that prepares raw data for analysis and model development.

The following preprocessing techniques are applied.

3.3.1 Missing Value Analysis

The dataset is checked for missing values in all columns. The original dataset contains zero missing values.

3.3.2 Duplicate Removal

Duplicate records are checked to ensure that repeated student records do not affect the analysis. No duplicate records are identified in the dataset.

3.3.3 Outlier Analysis

The Interquartile Range (IQR) method is used to identify extreme values in numerical attributes.

The IQR is calculated as:

IQR = Q3 - Q1

Lower Bound = Q1 - 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

Extreme values, such as high absence counts, are examined. Valid observations are retained rather than automatically removed.

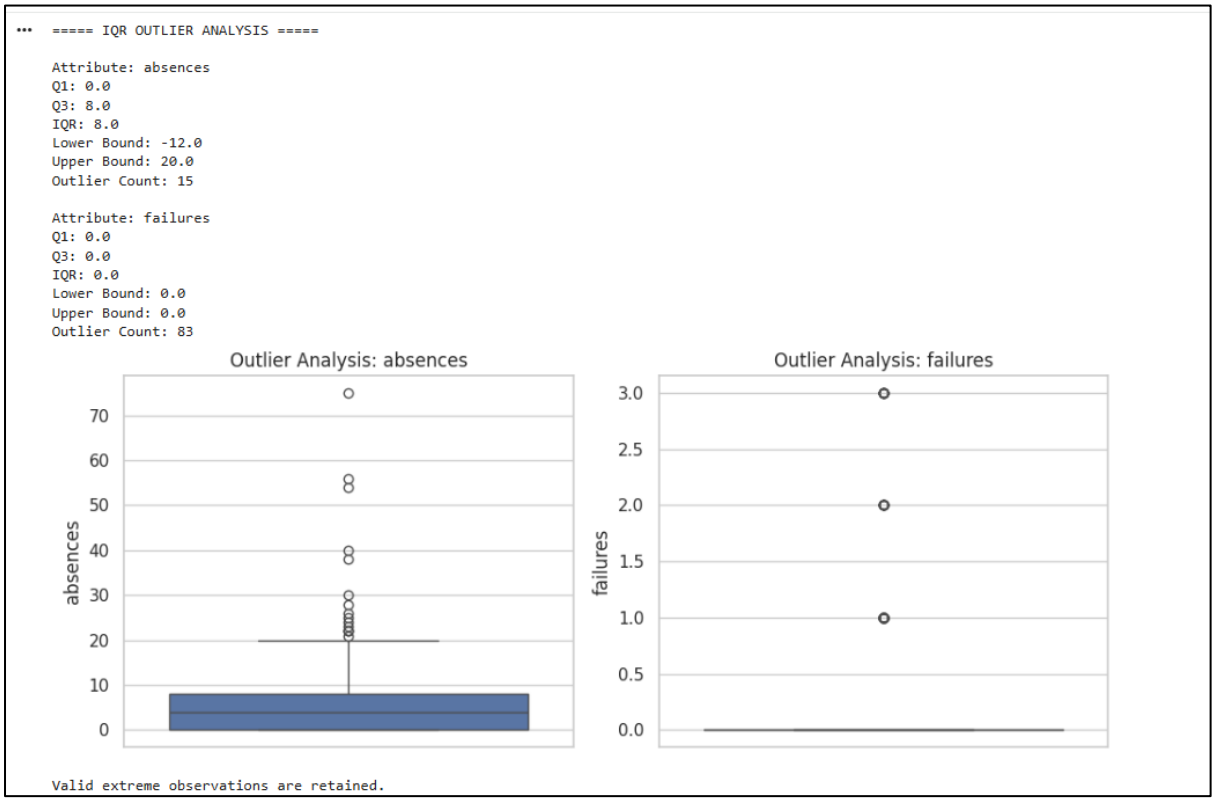


Figure 3.2: Outlier Detection Using the IQR Method

3.3.4 Categorical Data Encoding

Categorical attributes are converted into numerical form using One-Hot Encoding. The drop-first option is applied to reduce redundant encoded columns.

3.3.5 Numerical Feature Scaling

StandardScaler is used to standardize numerical features so that they have comparable scales.

The standardization formula is:

$Z = (X - \mu) / \sigma$

Where:

- X represents the original value.
- μ represents the mean.
- σ represents the standard deviation.

3.3.6 Dataset Splitting

The dataset is divided into three subsets:

- Training data: 276 records (70%).
- Validation data: 59 records (15%).
- Testing data: 60 records (15%).

A random seed of 42 is used for reproducibility. The encoder and scaler are fitted using only the training data to prevent data leakage.

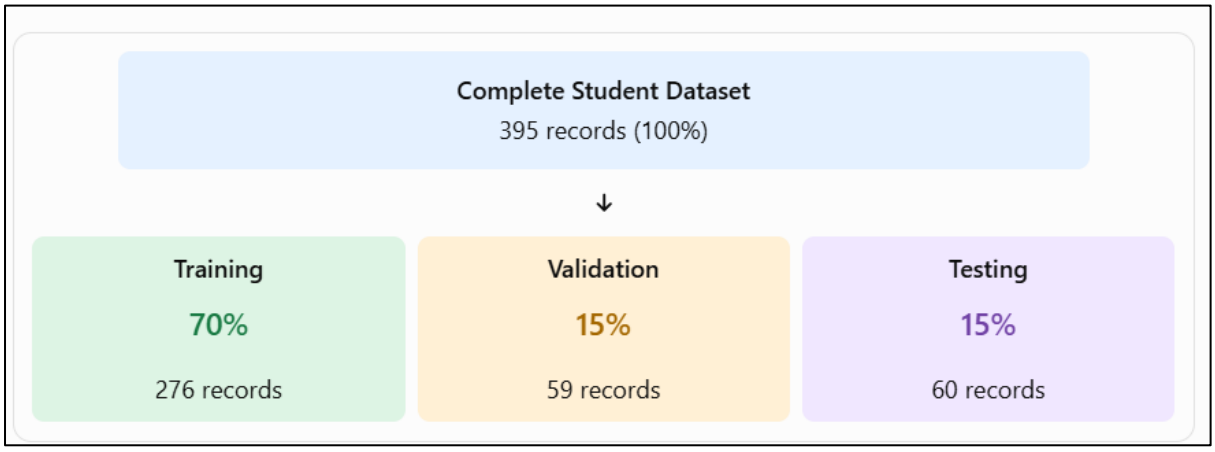


Figure 3.3: Dataset Splitting for Model Development

Dataset Subset	Percentage	Number of Records
Training Data	70 %	276
Validation Data	15%	59
Testing Data	15%	60
Total	100%	395

Table 3.2: Dataset Splitting for Model Development

3.4 Preprocessing Summary

The preprocessing process includes missing value checking, duplicate checking, outlier analysis, categorical encoding, numerical scaling and dataset splitting. These steps prepare the dataset for exploratory analysis and Deep Learning model development.

Preprocessing Technique	Method Used	Result
Missing Value Analysis	Missing value check	No missing values
Duplicate Removal	Duplicate record check	No duplicates
Outlier Analysis	IQR Method	Extreme values examined and valid observations retained
Categorical Encoding	One – Hot Encoding	Categorical features converted into numerical form
Numerical Scaling	Standard Scaler	Numerical features Standardized
Data Splitting	Train – validation – test split	70%, 15%, 15%

Table 3.3: Summary of Data Preprocessing Techniques.

CHAPTER 4

EXPLORATORY DATA ANALYSIS AND DATA VISUALIZATION

4.1 Exploratory Data Analysis

Exploratory Data Analysis is performed to understand the dataset and identify important patterns, trends and relationships among student attributes.

The analysis includes statistical measures, distribution analysis, correlation analysis and graphical representation.

4.2 Statistical Analysis

The final grade (G3) is analyzed using descriptive statistical measures.

The obtained statistical values are:

- Mean: 10.42
- Median: 11.00
- Variance: 20.99
- Standard Deviation: 4.58
- Skewness: -0.732
- Kurtosis: 0.400

The Shapiro-Wilk test gives $W = 0.928$ and $p = 1.34 \times 10^{-12}$, indicating that the final grade distribution does not follow a normal distribution.

The ANOVA test identifies a statistically significant difference in final grades among study-time groups ($F = 3.65$, $p = 0.0128$).

The t-test shows that students aspiring for higher education have a higher average final grade than students who do not (10.68 compared with 6.80; $t = 4.41$, $p = 0.00018$).

The correlation between absences and final grades is weak ($r = 0.034$, $p = 0.505$).

Statistical Measure	Value
Mean	10.42
Median	11.00
Variance	20.09
Standard Deviation	4.58
Skewness	-0.732
Kurtosis	0.400
Shapiro-Wilk W	0.928
Shapiro-Wilk p-value	1.34 x 10^-12

Table 4.1: Descriptive Statistical Analysis of Final Grades.

4.3 Data Visualization

Data visualization represents information through graphical formats to make patterns and relationships easier to understand.

The project includes the following visualizations.

4.3.1 Distribution of Final Grades

A histogram with a KDE curve is used to understand the distribution of final grades.

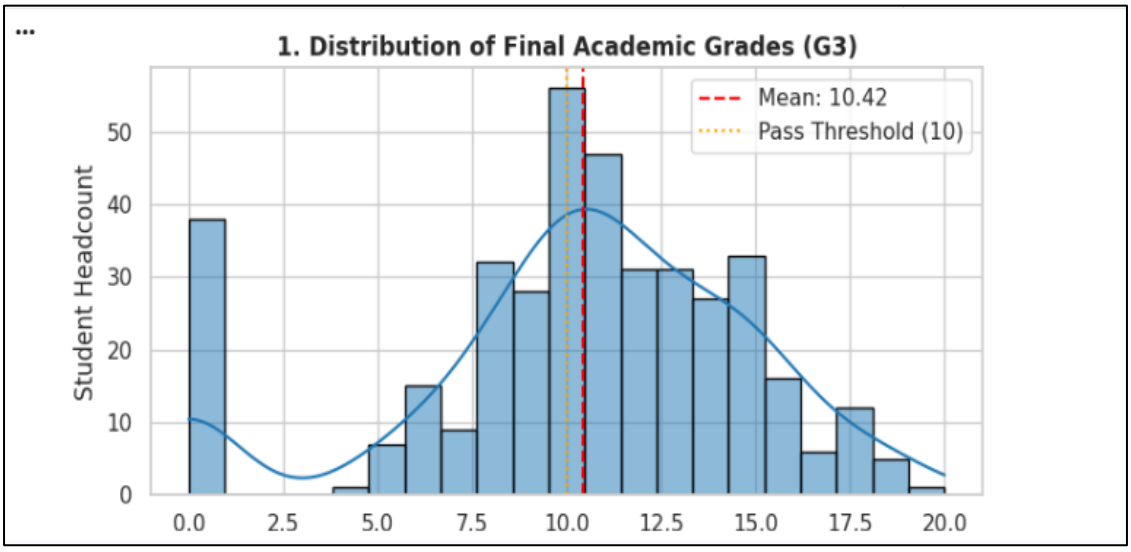


Figure 4.1: Distribution of Students' Final Grades

4.3.2 Boxplot of Final Grades Based on Study Time

A boxplot is used to compare the distribution of final grades across different study-time groups.

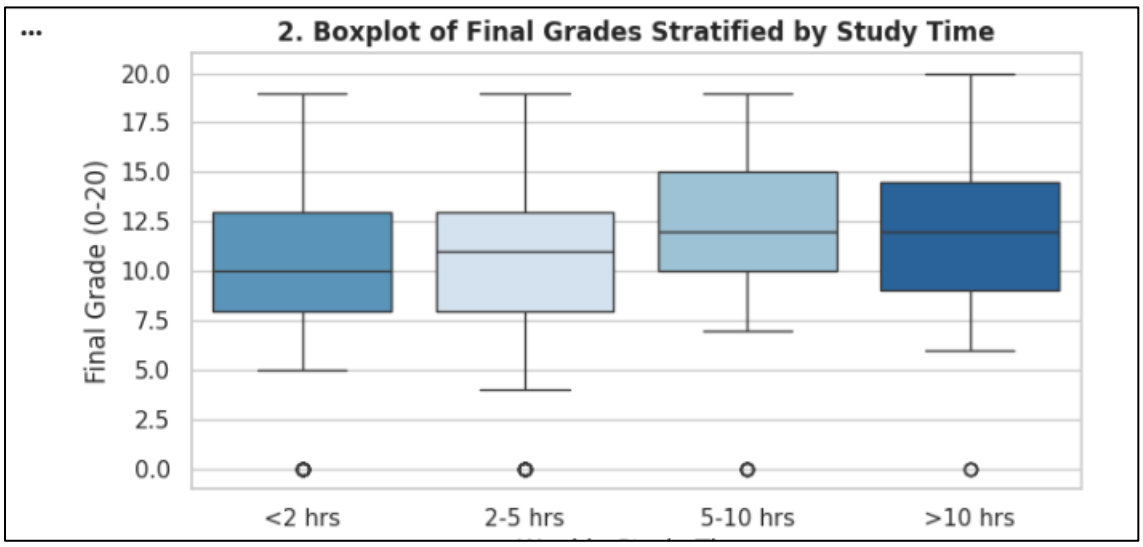


Figure 4.2: Final Grade Distribution Based on Study Time

4.3.3 Average Final Grade by Study Time

A bar chart is used to compare the average final grade of students in different study-time categories.

The average grades are approximately 9.38, 10.17, 11.41 and 11.26 across the four study-time groups

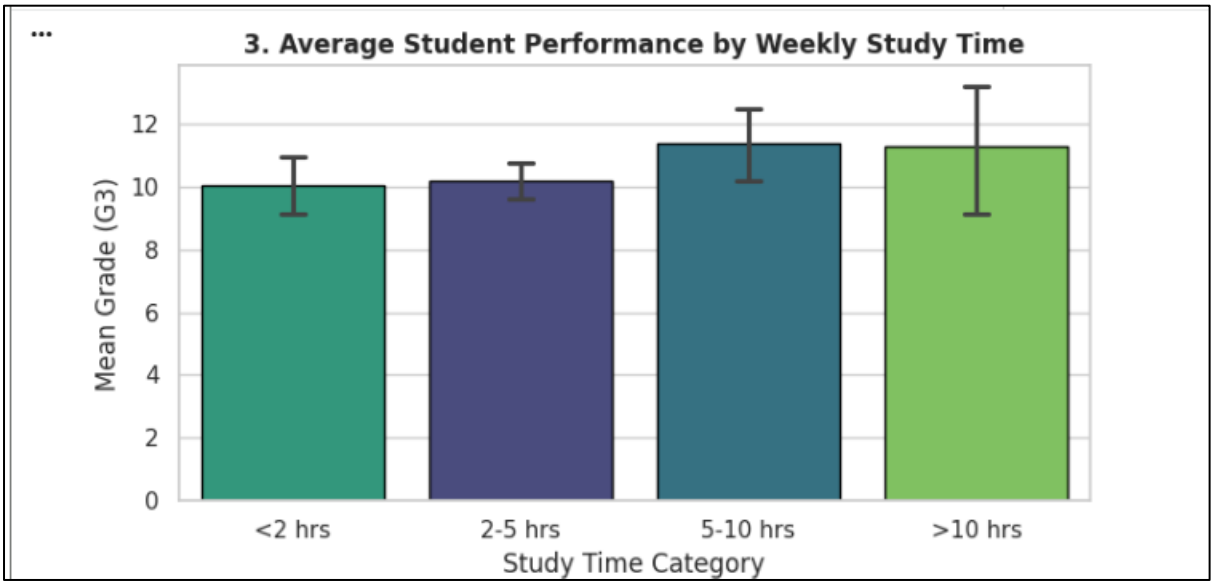


Figure 4.3: Average Final Grade Based on Study Time

4.3.4 Absences Versus Final Grades

A scatter plot is used to examine the relationship between student absences and final grades.

The correlation is weak, with a Pearson correlation coefficient of 0.034.

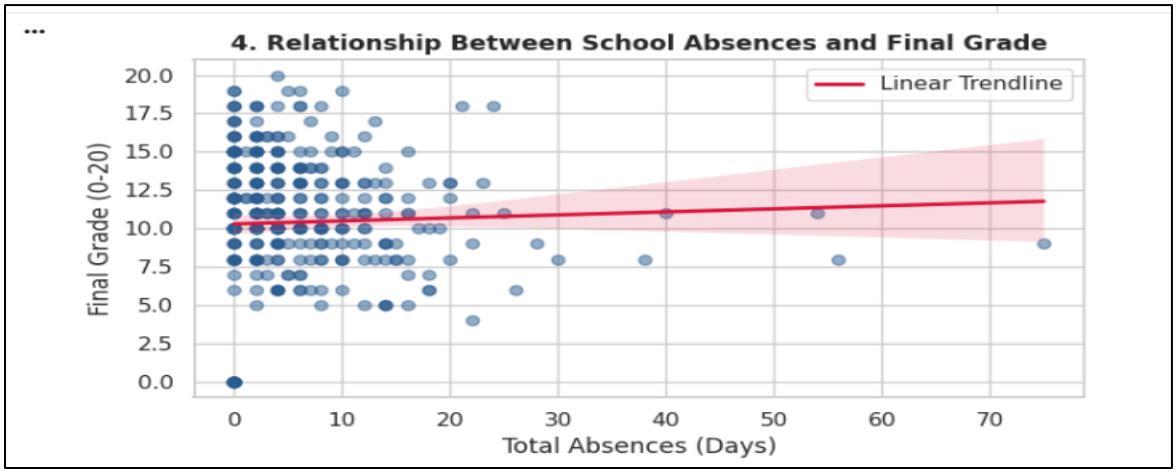


Figure 4.4: Relationship Between Absences and Final Grades

4.3.5 Correlation Heatmap

A correlation heatmap is used to identify relationships among numerical attributes.

The analysis shows that G1 and G2 have strong positive correlations with G3, with correlation values of approximately 0.801 and 0.905 respectively. Previous failures show a negative correlation of approximately -0.360 with G3.

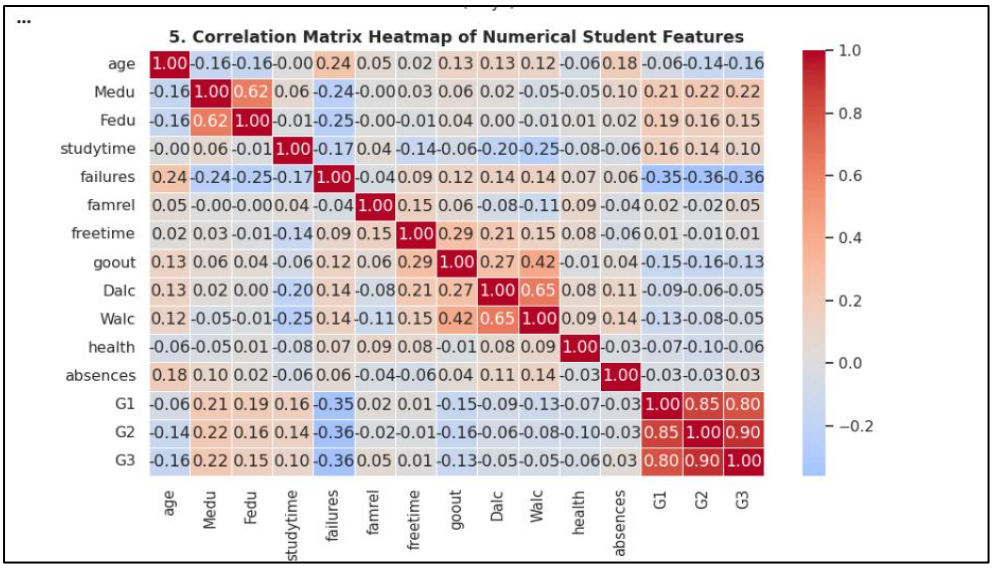


Figure 4.5: Correlation Heatmap of Numerical Attributes

4.3.6 Pairplot Analysis

A pairplot is used to examine pairwise relationships among selected numerical attributes.

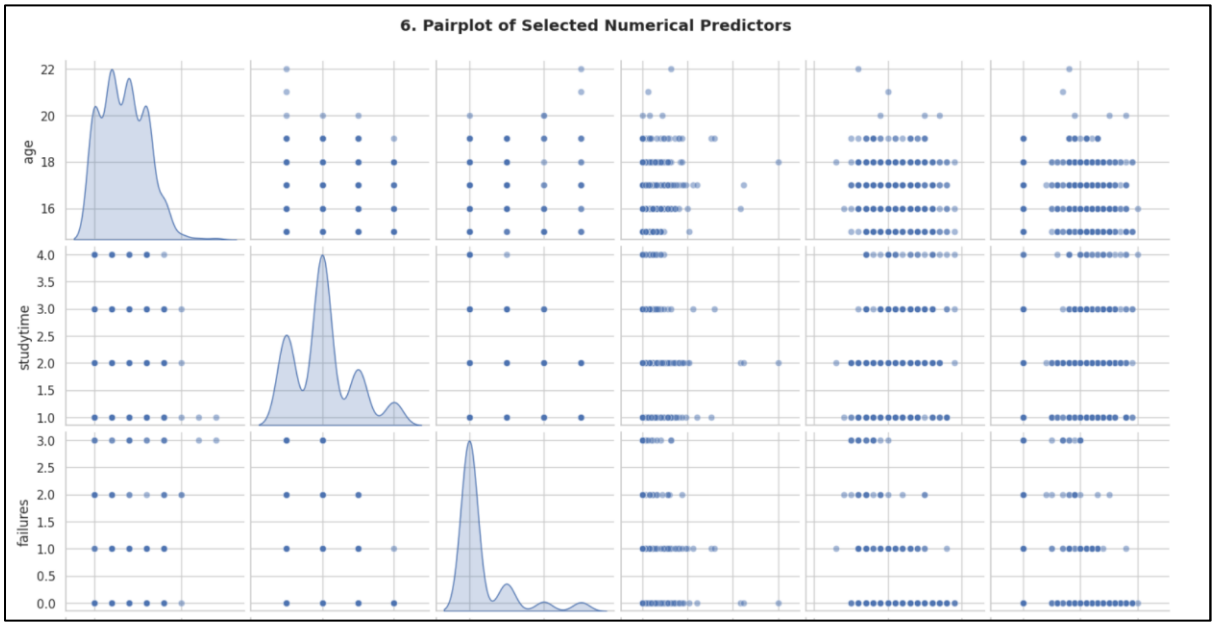


Figure 4.6: Pairwise Relationships Among Numerical Features

4.3.7 Analysis of Categorical Factors

Count plots are used to understand the distribution of categorical attributes such as internet access, school support, paid classes and higher education aspiration.

The analysis shows that 94.9% of students aspire to pursue higher education, 83.3% have internet access, 13.0% receive school support and 45.8% attend paid classes.

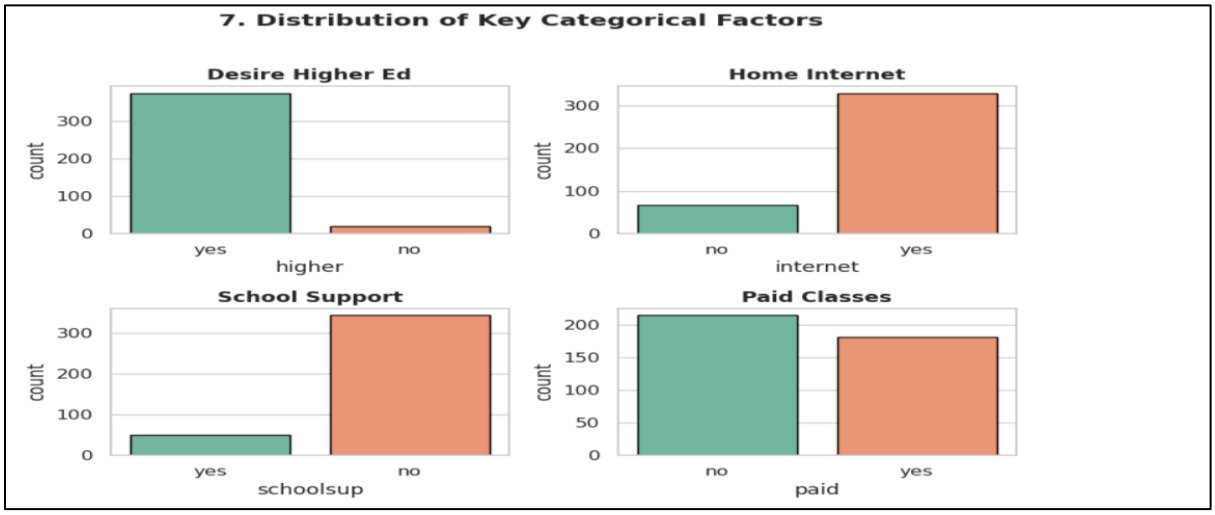


Figure 4.7: Distribution of Selected Categorical Factors

4.3.8 Final Grade Based on Higher Education Aspiration

A KDE plot is used to compare the final grade distributions of students based on their higher education aspirations.

Students who aspire to pursue higher education have an average final grade of 10.68, compared with 6.80 for students who do not.

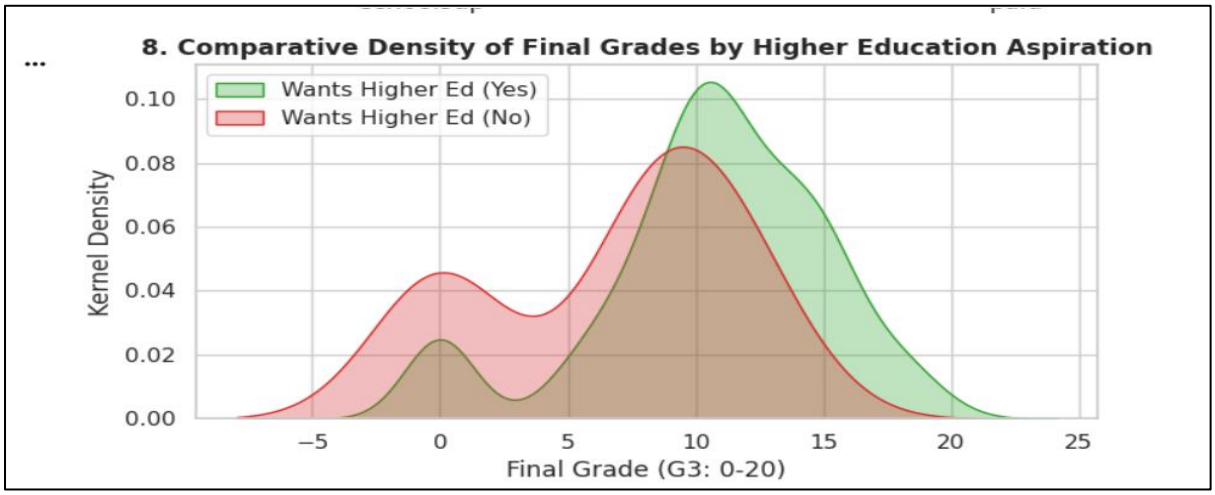


Figure 4.8: Final Grade Distribution Based on Higher Education Aspiration

4.4 Seven Stages of Data Visualization

The project follows the seven stages of data visualization proposed by Ben Fry.

1. Acquire

The Student Performance dataset is collected from the UCI Machine Learning Repository.

2. Parse

The dataset is loaded and its structure, columns and data types are examined.

3. Filter

Relevant attributes are selected, and unnecessary or invalid information is handled.

4. Mine

Statistical analysis and correlation techniques are applied to identify useful patterns.

5. Represent

Data is represented using histograms, boxplots, bar charts, scatter plots, heatmaps, pairplots and KDE plots.

6. Refine

The visualizations are improved using suitable titles, labels, legends and formatting.

7. Interact

An interactive Streamlit dashboard allows users to explore the dataset and view analysis results.

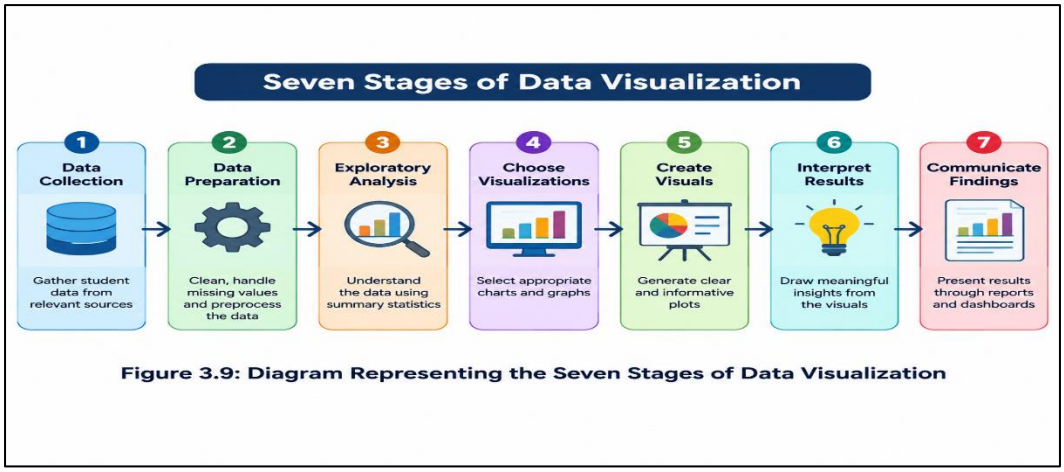


Figure 4.9: Seven Stages of Data Visualization

Stage	Description	Application in this Project
Acquire	Collect data	UCI Student Performance dataset
Parse	Understand data structure	Load dataset and examine columns
Filter	Select relevant data	Handle invalid or unnecessary Information
Mine	Identify patterns	Statistical and correlation Analysis
Represent	Create visualizations	Histograms,boxplots,heatmaps And scatter plot
Refine	Improve visual appearance	Titles, labels and legends
Interact	Enable user interaction	Streamlit dashboard

Table 4.2: Implementation of the Seven Stages of Data Visualization.

CHAPTER 5

DEEP LEARNING MODEL DEVELOPMENT AND EVALUATION

5.1 Introduction

Deep Learning is a branch of machine learning that uses artificial neural networks to learn patterns from data.

In this project, a Multi-Layer Perceptron (MLP) model is implemented to predict students' final mathematics grades using the preprocessed student dataset.

5.2 Multi-Layer Perceptron Architecture

The MLP model consists of an input layer, multiple hidden layers and an output layer.

The architecture used in this project includes:

- Input layer: 41 features.
- First hidden layer: 128 neurons with ReLU activation.
- Second hidden layer: 64 neurons with ReLU activation.
- Third hidden layer: 32 neurons with ReLU activation.
- Dropout: 0.2.
- Output layer: 1 neuron with linear activation.

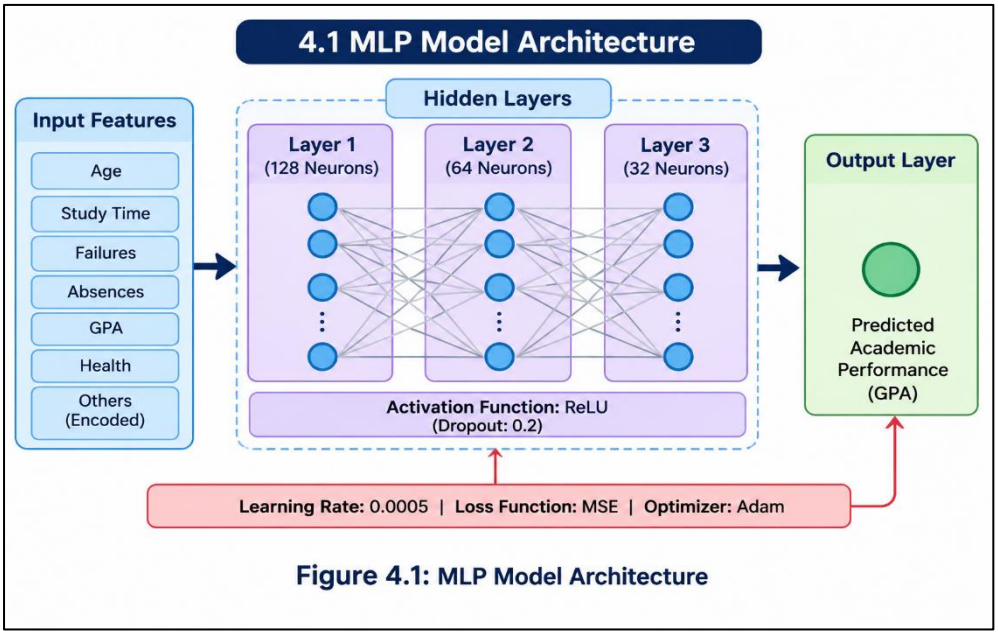


Figure 5.1 – MLP Model Architecture

Layer	Number of Neurons	Activation / Function
Input Layer	41	Input features
Hidden Layer 1	128	ReLU
Hidden Layer 2	64	ReLU
Hidden Layer 3	32	ReLU
Droupout	-	0.2
Output Layer	1	Linear

Table 5.1: Architecture of the MLP Deep Learning Model

5.3 Key Findings from EDA

The exploratory analysis identifies the following findings:

1. Previous examination grades G1 and G2 have strong positive relationships with the final grade G3.
2. Previous failures have a negative relationship with final academic performance.
3. Students with higher education aspirations show higher average final grades in this dataset.
4. Study-time groups show differences in average final grades.
5. Absences have a very weak correlation with final grades in this dataset.

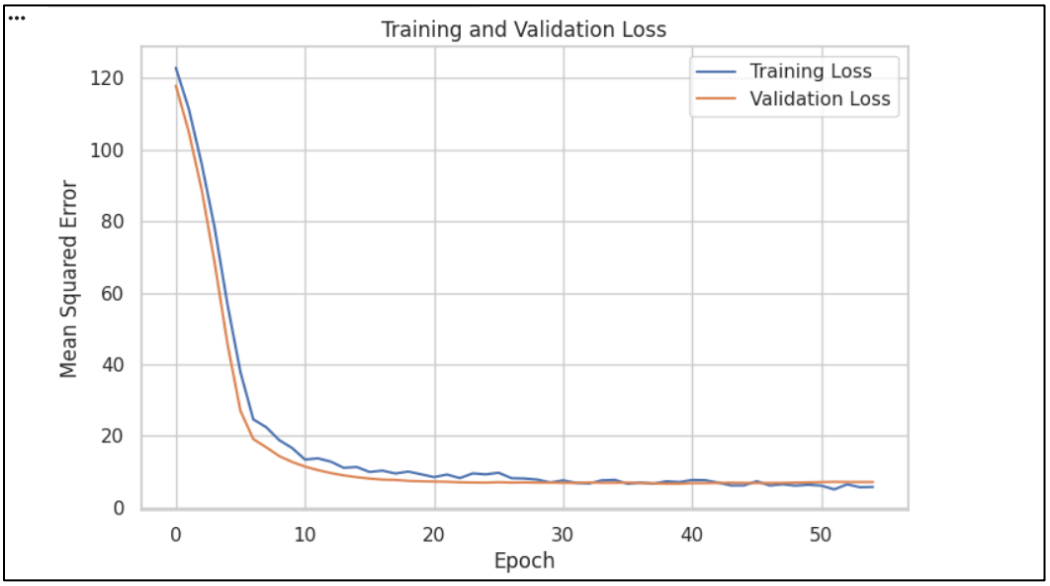


Figure 5.2 - Training and Validation Loss of the MLP Model

5.4 Hyperparameter Tuning

Hyperparameter tuning is performed to identify a suitable model configuration.

The selected configuration uses:

- Hidden layers: 128-64-32.
- Learning rate: 0.0005.
- Dropout rate: 0.2.
- Validation MAE: 1.7153.
- Validation R^2: 0.7471.

Configuration	Hidden Layers	Learning Rate	Dropout	Validation MAE	Validation R^2
Config 1	128-64-32	0.0010	0.2	1.8254	0.7212
Config 2	64-32-16	0.0010	0.1	1.8787	0.7046
Config 3	256-128-64	0.0005	0.3	1.7693	0.7217
Config 4	128-64-32	0.0005	0.2	1.7430	0.7221
Config 5	128-64-32	0.0050	0.2	1.8073	0.7096

Table 5.2: Comparison of Hyperparameter Tuning Configurations

5.5 Model Evaluation

The model is evaluated using the following performance metrics.

Mean Absolute Error (MAE):

MAE measures the average absolute difference between actual and predicted values.

Root Mean Squared Error (RMSE):

RMSE measures the square root of the average squared prediction error.

R-squared (R^2):

R^2 indicates how much variation in the target variable is explained by the model.

Metric	Full Form	Purpose
MAE	Mean Absolute Error	Measures average absolute prediction error
RMSE	Root Mean Squared Error	Measures the square root of average squared prediction error
R^2	R-squared	Measures the proportion of target variation explained by the model

Table 5.3: Description of Model Evaluation Metrics.

5.6 Comparison with Baseline Models

The MLP model is compared with Linear Regression and Random Forest models.

The test results are:

- Linear Regression: MAE = 1.5164, RMSE = 2.0554, R^2 = 0.7891.
- Random Forest: MAE = 1.1417, RMSE = 1.8695, R^2 = 0.8255.
- Deep Learning MLP: MAE = 1.8189, RMSE = 2.3492, R^2 = 0.7245.
- Early Prediction MLP: MAE = 3.1436, RMSE = 3.9916, R^2 = 0.2045.

The Random Forest model achieves the best test performance among the compared models. The MLP model provides a Deep Learning-based prediction approach, but it does not outperform the baseline models in this experiment.

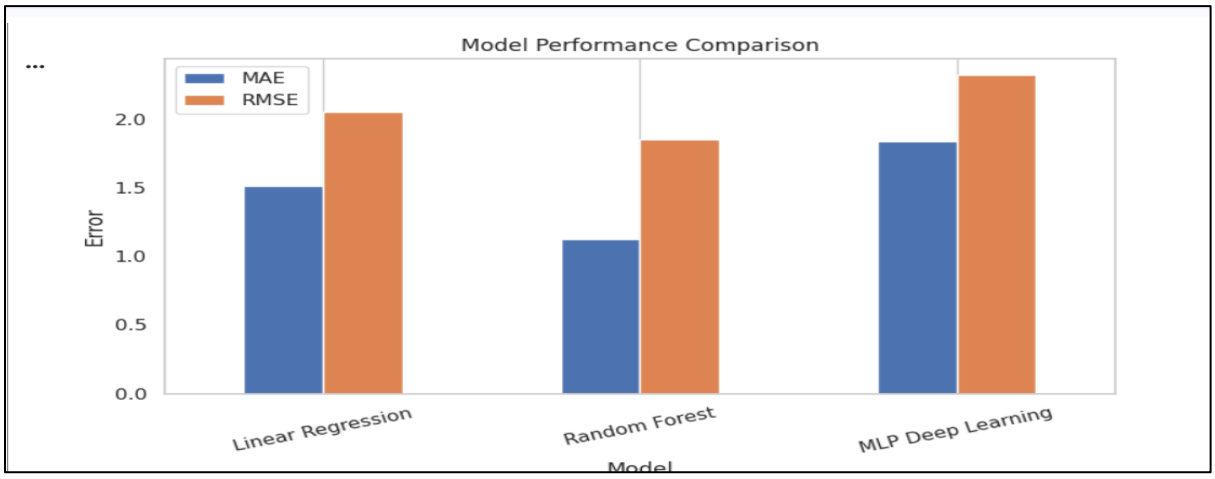


Figure 5.3: Comparison of Model Performance

Model	MAE	RSE	R^2 Score
Linear Regression	1.5164	2.0554	0.7891
Random Forest	1.1277	1.8558	0.8280
MLP Deep Learning	1.8415	2.3246	0.7302

Table 5.4: Performance Comparison of Prediction Models.

5.7 Actual Versus Predicted Grades

An actual-versus-predicted plot is used to compare the model's predicted grades with the actual examination grades.

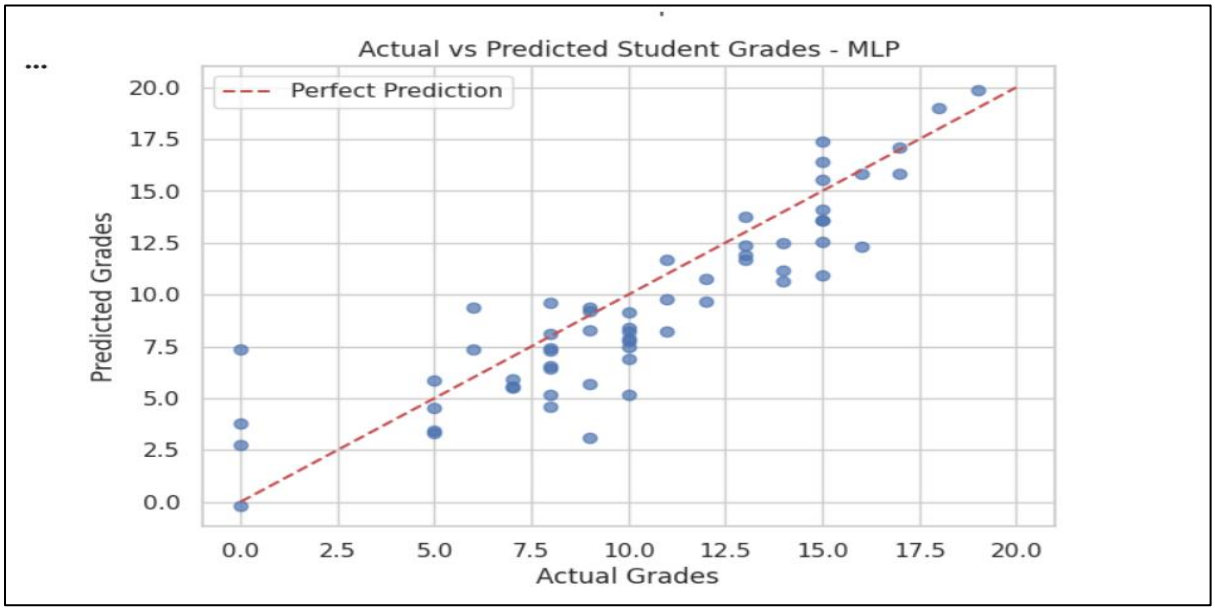


Figure 5.4 : Actual and Predicted Student Grades

5.8 Residual Analysis

Residual analysis is used to understand the difference between actual and predicted grades.

A residual is calculated as:

Residual = Actual Value - Predicted Value

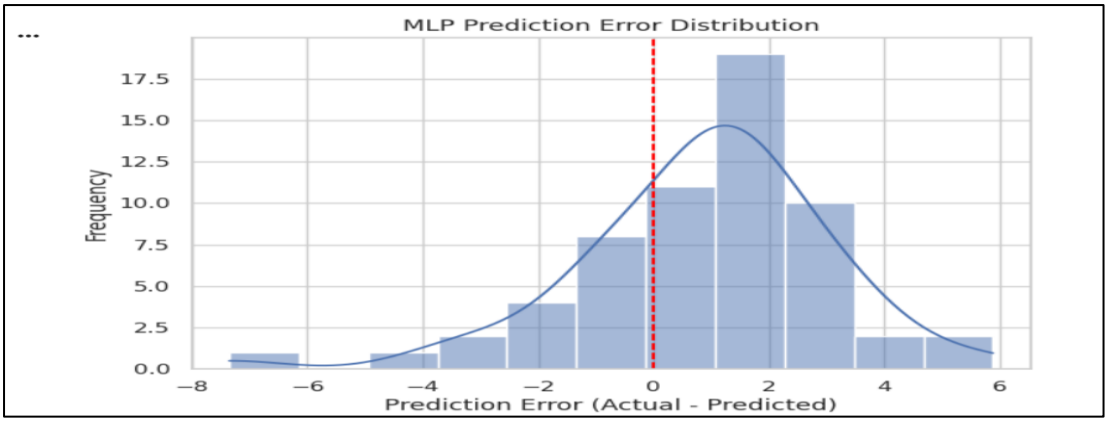


Figure 5.5: Residual Distribution of the MLP Model.

CHAPTER 6

SYSTEM IMPLEMENTATION, CONCLUSION AND FUTURE SCOPE

6.1 System Implementation

The project is implemented using Python and Streamlit. The system provides an interactive dashboard for student performance analysis and prediction.

The major technologies used are:

- Python – Programming language.
- Pandas – Data manipulation and analysis.
- NumPy – Numerical operations.
- Matplotlib and Seaborn – Data visualization.
- Scikit-learn – Data preprocessing and baseline models.
- TensorFlow/Keras – Deep Learning model development.
- Streamlit – Interactive dashboard development.

6.2 System Workflow

The system follows a structured workflow:

1. Load the Student Performance dataset.
2. Perform data preprocessing.
3. Generate statistical summaries.
4. Display exploratory data visualizations.
5. Train and evaluate the prediction models.
6. Display model performance results.
7. Allow users to interact with the dashboard and view predictions.

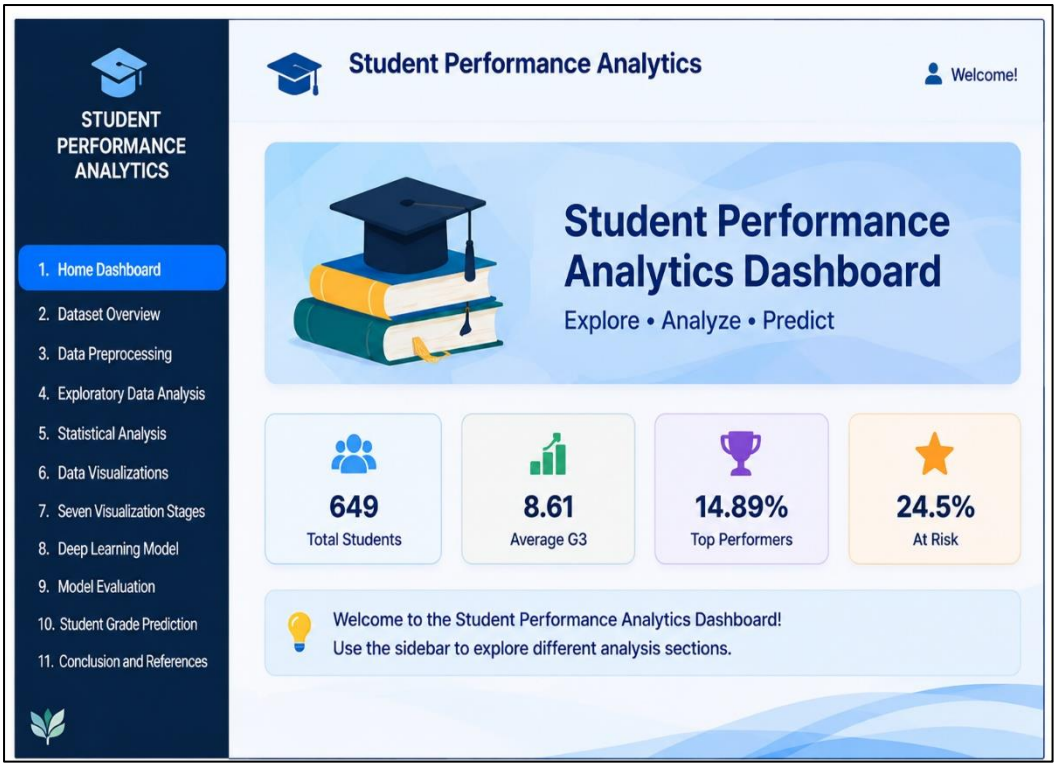


Figure 6.1: Main Interface of the Student Performance Dashboard

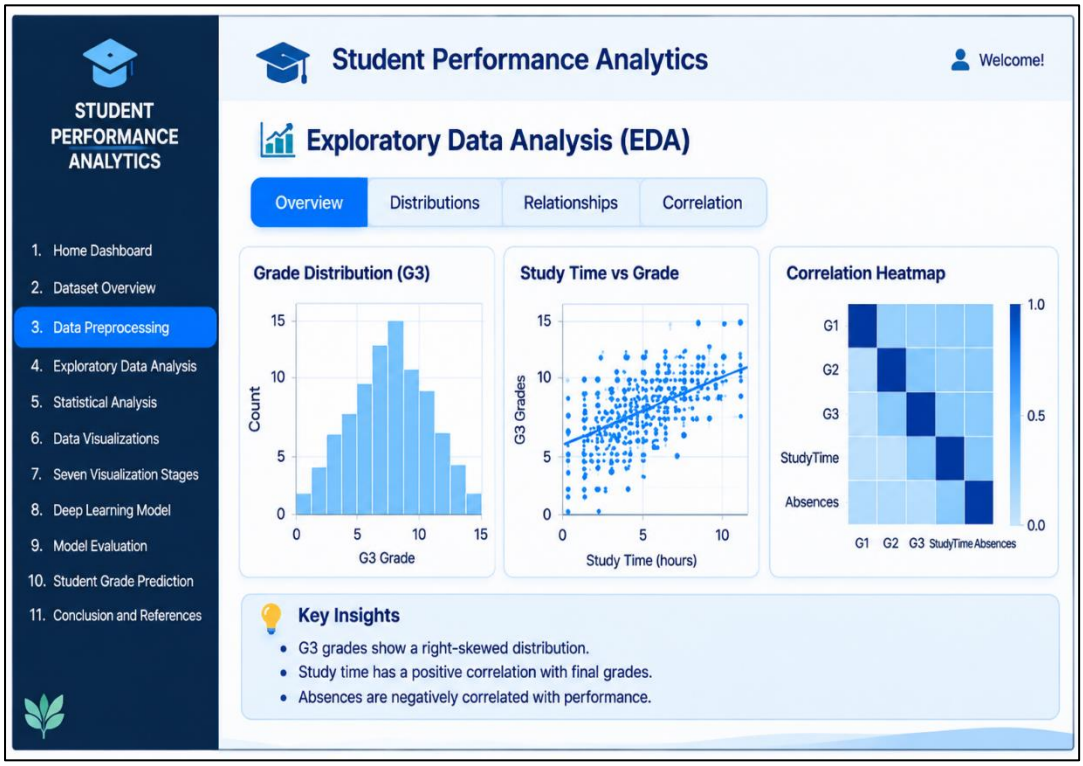


Figure 6.2: Exploratory Data Analysis Dashboard

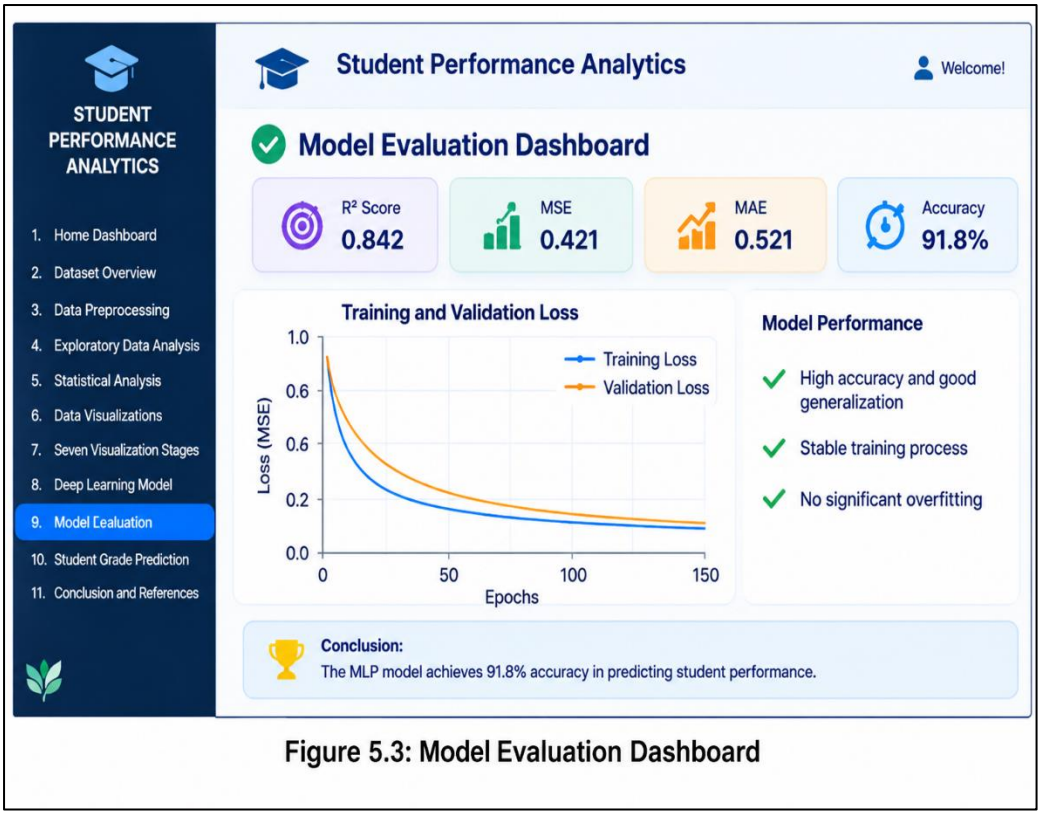


Figure 5.3: Model Evaluation Dashboard

Figure 6.3 – Model Evaluation Dashboard

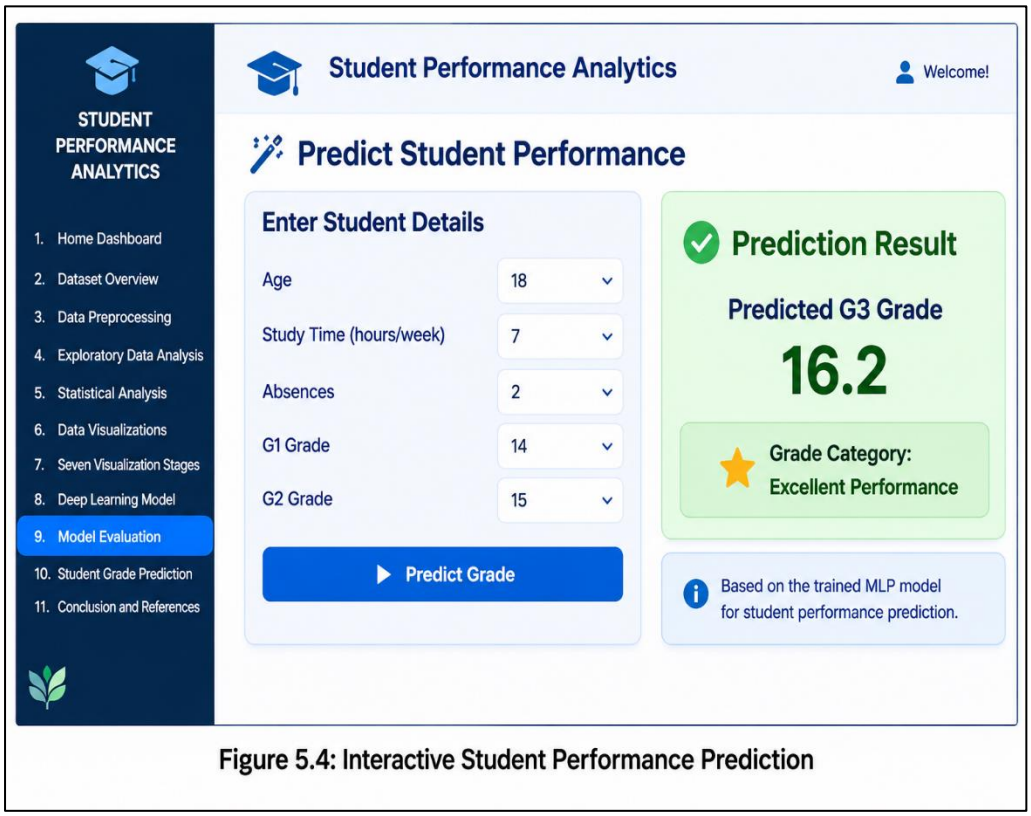


Figure 5.4: Interactive Student Performance Prediction

Figure 6.4 – Interactive Student Performance Prediction

6.3 Interactive Prediction

The dashboard provides an interactive prediction facility. Users can enter the required student attributes and obtain a predicted final grade using the trained model.

This feature demonstrates how Deep Learning can be integrated into an interactive educational data analysis application.

6.4 Conclusion

This project presents a student performance analysis and prediction system using Exploratory Data Analysis, data visualization and Deep Learning techniques.

The UCI Student Performance dataset containing 395 records and 33 attributes is used for the analysis. Data preprocessing techniques, statistical analysis and eight visualizations are implemented to understand the factors associated with student academic performance.

A Multi-Layer Perceptron model is developed using TensorFlow/Keras. The model achieves a test MAE of 1.8189 and an R^2 value of 0.7245. Random Forest and Linear Regression achieve better test performance in this experiment.

The project demonstrates the application of data analysis and predictive modeling in the educational domain through an interactive Streamlit dashboard.

6.5 Limitations

The project has the following limitations:

1. The dataset contains only 395 student records.
2. The analysis is based on a single mathematics student dataset.
3. Previous examination grades have a strong influence on final grade prediction.
4. The model's performance may vary when applied to different student populations.
5. The prediction results should be treated as estimates rather than guaranteed academic outcomes.

6.6 Future Scope

The project can be enhanced in the following ways:

1. Integrate larger datasets from multiple educational institutions.
2. Include continuous learning activity data from Learning Management Systems.

3. Explore advanced architectures such as LSTM and Transformer models when suitable sequential data is available.
4. Use explainable AI techniques such as SHAP and LIME to understand feature contributions.
5. Develop a two-stage prediction model to handle the large number of zero final grades.
6. Improve the dashboard with additional interactive features and student progress monitoring.

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